

Unifying Special Relativity and Quantum Mechanics

Lecture 9 — The Massive Vector Field: Polarization Vectors, Transversality, the Polarization Sum, and Spin–Statistics from Microcausality

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as Lectures 1–8: everything in the normal text and in the yellow boxes is a faithful transcription of the handwritten notes (yellow boxes correspond to the results that were highlighted or framed in the original, whether in yellow, pink/red, or plain frames); everything inside a blue box like this one is extra material that I added.

Pagination of this scan. The notes open yet another chapter and restart the page numbering from 1 a fourth time; the scanned sheets carry the circled numbers 3–18, so the markers below are written [p. 3****], ..., [p. 18****] to keep them apart from the earlier restarts. *Note that pages 1–2 of this chapter are not present in the scan:* the document therefore starts mid-discussion, with the closing remark about the “most economical” choice of Lorentz representation. Judging from the context, the missing pages contained the beginning of the treatment of the spin-1 field (the choice $D^{(1/2,1/2)}$ and its motivation via the selection rule of Lecture 8).

Where we left off (recap of Lecture 8). The consistency (intertwining) conditions on the coefficient functions were reduced to a two-step procedure — solve at zero momentum using the little group, then boost — and the finite-dimensional representations of the Lorentz group were classified by pairs (j_1, j_2) , with the selection rule: to describe spin s , the decomposition of $D^{(j_1, j_2)}$ under rotations, $\bigoplus_{j=|j_1-j_2|}^{j_1+j_2} D_j$, must contain $j = s$. For $s = 1$ the smallest admissible choice is $(\frac{1}{2}, \frac{1}{2}) = 0 \oplus 1$, i.e. the four-vector. This lecture carries out the whole programme explicitly for that case.

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1 The four-vector field and the choice $D^{(1/2,1/2)}$ [p. 3****]

The choice $D^{(1/2,1/2)}(\Lambda)$ is the “most economical” one.

↪ Other choices are possible, at the price of introducing additional constraints to eliminate unphysical degrees of freedom. While such formulations can describe the same free particle, they may differ at the level of interactions, thus providing a phenomenological probe of different implementations.

Let us now try to make contact with more “familiar” Lorentz representations. Consider a four-vector field $V^\mu(x)$. It transforms under Lorentz according to the defining representation:

$$V^\mu(x) \longrightarrow \Lambda^\mu{}_\nu V^\nu(\Lambda^{-1}x)$$

► This is an irreducible representation, it must be somewhere in our classification. The correct identification is

$$V^\mu \sim D^{(1/2,1/2)}$$

[p. 4****] 1) $V^\mu(x)$ has four real components, it must transform according to a four-dimensional real representation.

2) If we write V^μ in components, we see that

$$V^\mu(x) = \begin{pmatrix} V^0(x) \\ \vec{V}(x) \end{pmatrix} \quad \begin{cases} V^0(x) : \text{does not transform under rotations,} \\ \vec{V}(x) : \text{does transform under rotations like a three-dimensional} \\ \text{vector } (j=1). \end{cases}$$

Consequently, $D^{(1/2,1/2)}$ has the right content under rotations to act properly on $V^\mu(x)$.

We thus write the field in four-vector form:

$$V^\mu(x) = \sum_{\sigma} \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\kappa u^\mu(\vec{p}, \sigma) e^{-ip \cdot x} a(\vec{p}, \sigma) + \eta v^\mu(\vec{p}, \sigma) e^{ip \cdot x} a^\dagger(\vec{p}, \sigma) \right]$$

instead of “ A ” we use the four-vector index “ μ ”;

we combine V_+ and V_- with generic coefficients;

we do not make assumptions about the statistics, but we leave open the possibility that $a, a^\dagger$ either commute or anti-commute.

Added explanation (not in the original notes)

Why $V^\mu \sim (\frac{1}{2}, \frac{1}{2})$ is forced. The two arguments in the notes are a dimension count and a rotation-content count, and together they pin the identification down: among the (j_1, j_2) list of Lecture 8, the four-dimensional ones are $(\frac{1}{2}, \frac{1}{2})$ and $(\frac{3}{2}, 0), (0, \frac{3}{2})$; only the first decomposes under rotations as $j = 0 \oplus 1$ — matching the split of V^μ into the rotation-scalar V^0 and the

rotation-vector $\vec{V}$ — and only the first is self-conjugate ($j_1 = j_2$), hence expressible in *real* form (Lecture 8, [p. 42***]), as a real four-vector field requires. The price of the economy is already visible: the representation carries a $j = 0$ piece that the spin-1 particle does not need. That surplus component is exactly what the transversality condition derived on [p. 7****] will remove. **On leaving κ, η and the statistics open.** This is the whole point of the lecture's second half: rather than assuming the spin-statistics rule (as Lecture 6 did), the notes keep both options alive and let *microcausality* decide — the promise made in the Lecture 7 blue box is redeemed on [p. 18****].

2 Solving the particle–field connection: zero momentum [p. 5****]

We follow our two-steps strategy.

1) **Coefficient functions at zero momentum.** The key equations are

$$D(\theta, \hat{n})^A{}_B u^B(\vec{0}, \sigma) = \sum_{\sigma'} (D_s(\theta, \hat{n}))_{\sigma'\sigma} u^A(\vec{0}, \sigma'),$$

$$D(\theta, \hat{n})^A{}_B v^B(\vec{0}, \sigma) = \sum_{\sigma'} (D_s(\theta, \hat{n}))_{\sigma'\sigma}^* v^A(\vec{0}, \sigma').$$

We set $s = 1$, and specify the representation D :

$$\Lambda_R(\theta, \hat{n})^\mu{}_\nu u^\nu(\vec{0}, \sigma) = \sum_{\sigma'} (D_1(\theta, \hat{n}))_{\sigma'\sigma} u^\mu(\vec{0}, \sigma')$$

$$\Lambda_R(\theta, \hat{n})^\mu{}_\nu v^\nu(\vec{0}, \sigma) = \sum_{\sigma'} (D_1(\theta, \hat{n}))_{\sigma'\sigma}^* v^\mu(\vec{0}, \sigma')$$

We consider infinitesimal rotations:

$$\begin{cases} \Lambda_R(\theta, \hat{n}) = \mathbb{1} - i\theta \hat{n} \cdot \vec{J}_{4\text{vec}} \\ D_1(\theta, \hat{n}) = \mathbb{1} - i\theta \hat{n} \cdot \vec{J}_{s=1} \end{cases}$$

$$(\vec{J}_{4\text{vec}})^\mu{}_\nu u^\nu(\vec{0}, \sigma) = \sum_{\sigma'} (\vec{J}_{s=1})_{\sigma'\sigma} u^\mu(\vec{0}, \sigma')$$

$$(\vec{J}_{4\text{vec}})^\mu{}_\nu v^\nu(\vec{0}, \sigma) = - \sum_{\sigma'} (\vec{J}_{s=1})_{\sigma'\sigma}^* v^\mu(\vec{0}, \sigma')$$

Added explanation (not in the original notes)

Where the extra minus sign comes from. Expanding the finite equations to first order in θ : on the left one gets $-i\theta(\hat{n} \cdot \vec{J}_{4\text{vec}})^\mu{}_\nu$; on the right, for u , $-i\theta(\hat{n} \cdot \vec{J}_{s=1})_{\sigma'\sigma}$ — the factors $-i\theta$ cancel. For v the right-hand side carries the *complex conjugate* matrix, $(\mathbb{1} - i\theta \hat{n} \cdot \vec{J}_{s=1})^* = \mathbb{1} + i\theta \hat{n} \cdot \vec{J}_{s=1}^*$, so relative to the left-hand side one is left with an overall minus sign, as boxed. Also, the zeroth-order terms give $\delta^\mu{}_\nu u^\nu = \sum_{\sigma'} \delta_{\sigma'\sigma} u^\mu$, an identity — so all the information sits at first order, which is why working infinitesimally loses nothing.

3 Observation: v from u , and the polarization vectors [p. 6****]

Observation. We compute explicitly $\vec{J}_{4\text{vec}}$, and we know that

$$(\vec{J}_{4\text{vec}})^* = -\vec{J}_{4\text{vec}}.$$

The equation for v^μ thus reads

$$(\vec{J}_{4\text{vec}}^*)^\mu{}_\nu v^\nu(\vec{0}, \sigma) = \sum_{\sigma'} (\vec{J}_{s=1})_{\sigma'\sigma}^* v^\mu(\vec{0}, \sigma')$$

$\rightsquigarrow$ if we solve the equation for $u^\mu(\vec{0}, \sigma)$, we automatically solve also the equation for $v^\mu(\vec{0}, \sigma)$ if we just take

$$v^\mu(\vec{0}, \sigma) = (u^\mu(\vec{0}, \sigma))^*.$$

We thus focus on

$$\begin{aligned} (\vec{J}_{4\text{vec}})^\mu{}_\nu \varepsilon^\nu(\vec{0}, \sigma) &= \sum_{\sigma'} (\vec{J}_{s=1})_{\sigma'\sigma} \varepsilon^\mu(\vec{0}, \sigma') \\ \varepsilon^\mu(\vec{0}, \sigma) &\equiv u^\mu(\vec{0}, \sigma) = (v^\mu(\vec{0}, \sigma))^* \end{aligned}$$

$\hookrightarrow$ *polarization vectors at zero momentum.*

Solution strategy. In the previous equation, “ μ ” is a free index, $\mu = 0, 1, 2, 3$, and also “ σ ” is a free index, $\sigma = \pm 1, 0$.

Added explanation (not in the original notes)

Why $(\vec{J}_{4\text{vec}})^* = -\vec{J}_{4\text{vec}}$. The rotation generators in the four-vector representation were computed in Lecture 2, [p. 37–38]: they are i times real antisymmetric matrices, e.g. $J_{4\text{vec}}^3 = iT^3$ with T^3 real, so $J_{4\text{vec}}^k$ is purely imaginary and conjugation flips its sign. (Equivalently: Λ_R is a *real* matrix, so $\Lambda_R = \Lambda_R^*$ forces $-i\vec{J}_{4\text{vec}} = (-i\vec{J}_{4\text{vec}})^* = +i\vec{J}_{4\text{vec}}^*$.) This little fact is what collapses two problems into one — and it is not an accident: it is the statement that the vector representation is self-conjugate, $(\frac{1}{2}, \frac{1}{2})^* \sim (\frac{1}{2}, \frac{1}{2})$ (Lecture 8, [p. 42****]). For representations with $j_1 \neq j_2$ (e.g. Weyl spinors) the two equations do *not* collapse, and u and v are genuinely different objects — the algebraic root of the particle/antiparticle distinction.

4 $\mu = 0$: transversality [p. 7****]

$\rightsquigarrow \mu = 0$:

$$\underbrace{(\vec{J}_{4\text{vec}})^0{}_\nu}_{=0} \varepsilon^\nu(\vec{0}, \sigma) = \sum_{\sigma'} (\vec{J}_{s=1})_{\sigma'\sigma} \varepsilon^0(\vec{0}, \sigma') \quad \Longrightarrow \quad \boxed{\sum_{\sigma'} (\vec{J}_{s=1})_{\sigma'\sigma} \varepsilon^0(\vec{0}, \sigma') = 0}$$

This equation implies

$$\varepsilon^0(\vec{0}, \sigma) = 0, \quad \sigma = 0, \pm 1.$$

This is an interesting result. Since $p_{\text{ref}}^\mu = (M, \vec{0})$, we get

$$p_{\text{ref}}^\mu \varepsilon_\mu(\vec{0}, \sigma) = 0 \quad \forall \sigma = 0, \pm 1.$$

▷ This is a scalar product in 4-dim; it remains true in any frame since it is Lorentz invariant:

$$p_\mu \varepsilon^\mu(\vec{p}, \sigma) = 0, \quad \sigma = 0, \pm 1 \quad \text{TRANSVERSALITY CONDITION}$$

In terms of the field $V^\mu(x)$, the transversality condition implies

$$\partial_\mu V^\mu(x) = 0$$

[p. 8****]

The free field $V^\mu(x)$, in addition to satisfying the KG equation, also verifies the additional constraint $\partial_\mu V^\mu = 0$ if we want to use it to describe massive spin-1 particles.

► This makes sense, since we only want to describe 3 physical d.o.f. and we only need 3 real field variables:

$$\underbrace{V^\mu(x)}_{\text{four real field components}} - \underbrace{1 \text{ constraint}}_{\partial_\mu V^\mu = 0} = 3 \text{ d.o.f.}$$

► This is only one equation (we contract “ μ ”); therefore, it can be used to eliminate only 1 d.o.f.

Added explanation (not in the original notes)

How the $\mu = 0$ equation kills ε^0 . The matrices $(\vec{J}_{s=1})_{\sigma'\sigma}$ are the spin-1 angular-momentum matrices of Lecture 4, [p. 10*]; the boxed condition says the three-component object $\varepsilon^0(\vec{0}, \cdot)$ is annihilated by *all three* of J^1, J^2, J^3 , i.e. it is a rotational *singlet* inside the spin-1 multiplet. Since D_1 is irreducible, the only such vector is zero (Schur). This is precisely the surplus $j = 0$ component of $(\frac{1}{2}, \frac{1}{2})$ being projected out, as anticipated: the field has four components, the particle has three states, and the mismatch is removed by one covariant condition.

Reading $\partial_\mu V^\mu = 0$. Acting with ∂_μ on the plane-wave expansion brings down $\mp i p_\mu$ contracted with ε^μ , which vanishes by transversality — so the field equation is a *consequence* of the particle content, not an extra postulate. Together with $(\square + M^2)V^\mu = 0$ (Lecture 8, valid for every free field), these are the *Proca equations* of the massive vector field; they are equivalent to the more familiar single equation $\partial_\mu F^{\mu\nu} + M^2 V^\nu = 0$ with $F^{\mu\nu} = \partial^\mu V^\nu - \partial^\nu V^\mu$, whose divergence gives $\partial_\nu V^\nu = 0$ automatically *provided* $M \neq 0$. That proviso matters: for $M = 0$ the constraint is not generated, and the extra freedom reappears as gauge invariance — which is how the massless case (the photon, with only 2 polarizations) escapes this counting.

5 $\mu = i$: the polarization vectors at zero momentum [p. 8****–9****]

↪ We now take $\mu = i$:

$$(J_{4\text{vec}}^k)^i{}_\nu \varepsilon^\nu(\vec{0}, \sigma) = \sum_{\sigma'} (J_{s=1}^k)_{\sigma'\sigma} \varepsilon^i(\vec{0}, \sigma') \quad (\text{since } \varepsilon^0 = 0)$$

$$(J_{4\text{vec}}^k)^i{}_j \varepsilon^j(\vec{0}, \sigma) = \sum_{\sigma'} (J_{s=1}^k)_{\sigma'\sigma} \varepsilon^i(\vec{0}, \sigma')$$

[p. 9****] We specify $k = 3$, for which

$$J_{s=1}^{k=3} = \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix} \quad \text{or, equivalently,} \quad (J_{s=1}^3)_{\sigma'\sigma} = \sigma \delta_{\sigma'\sigma}.$$

We thus get

$$(J_{4\text{vec}}^3)^i{}_j \varepsilon^j(\vec{0}, \sigma) = \sigma \varepsilon^i(\vec{0}, \sigma)$$

► This is an eigenvalue equation: $\vec{\varepsilon}(\vec{0}, \sigma)$ are eigenstates of

$$(J_{4\text{vec}}^3)^i{}_j = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with eigenvalues $\pm 1, 0$. These eigenstates are

$$\vec{\varepsilon}(\vec{0}, 1) = -\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}, \quad \vec{\varepsilon}(\vec{0}, -1) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \\ 0 \end{pmatrix}, \quad \vec{\varepsilon}(\vec{0}, 0) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

We thus find, at zero momentum,

$$\varepsilon^\mu(\vec{0}, 1) = -\frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ i \\ 0 \end{pmatrix}, \quad \varepsilon^\mu(\vec{0}, -1) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -i \\ 0 \end{pmatrix}, \quad \varepsilon^\mu(\vec{0}, 0) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

polarization vectors at zero momentum.

Added explanation (not in the original notes)

Checking the eigenvectors, and the meaning of the phases. With $(J_{4\text{vec}}^3)^i{}_j$ as above, $J^3(1, \pm i, 0)^T = (\mp i \cdot i, i \cdot 1, 0)^T \dots$ explicitly: $J^3(1, i, 0)^T = (-i \cdot i, i \cdot 1, 0)^T = (1, i, 0)^T$, eigenvalue $+1$; $J^3(1, -i, 0)^T = (+i \cdot i, i, 0)^T = -(1, -i, 0)^T$, eigenvalue -1 ; and $\hat{z}$ is annihilated, eigenvalue 0 . ✓ The combinations $\mp(\hat{x} \pm i\hat{y})/\sqrt{2}$ are the familiar *spherical basis* of vector calculus: the same change of basis, from Cartesian x, y, z to $m = +1, 0, -1$, that was mentioned in Lecture 4 ([p. 10*] blue box) as relating the $j = 1$ matrices to the ε_{kij} generators. The overall -1 for $\sigma = +1$ is the standard Condon–Shortley phase convention; it is pure convention, but it must be kept consistently (it cancels in the polarization sum computed below). Normalization: $\varepsilon(\vec{0}, \sigma) \cdot \varepsilon(\vec{0}, \sigma')^* = \delta_{\sigma\sigma'}$ in the three-dimensional sense, i.e. $\varepsilon^\mu \varepsilon_\mu^* = -1$ in the four-dimensional one (space-like unit vectors, as they must be since they are orthogonal to the time-like p_{ref}).

6 Coefficient functions at generic momentum [p. 10****–11****]

2) Coefficient functions at generic momentum. We boost the zero-momentum solutions. The relevant equations are

$$u^A(\vec{p}, \sigma) = D(\Lambda_B(\eta, \hat{p}))^A{}_B u^B(\vec{0}, \sigma), \quad v^A(\vec{p}, \sigma) = D(\Lambda_B(\eta, \hat{p}))^A{}_B v^B(\vec{0}, \sigma).$$

In our case we get

$$\begin{cases} u^\mu(\vec{p}, \sigma) = \Lambda_B(\eta, \hat{p})^\mu{}_\nu \varepsilon^\nu(\vec{0}, \sigma) \\ v^\mu(\vec{p}, \sigma) = \Lambda_B(\eta, \hat{p})^\mu{}_\nu (\varepsilon^\nu(\vec{0}, \sigma))^* \end{cases}$$

▷ Since Λ_B is a real matrix, we can compute the complex conjugate of this equation, and get

$$\begin{cases} u^\mu(\vec{p}, \sigma) = \Lambda_B(\eta, \hat{p})^\mu{}_\nu \varepsilon^\nu(\vec{0}, \sigma) \\ (v^\mu(\vec{p}, \sigma))^* = \Lambda_B(\eta, \hat{p})^\mu{}_\nu \varepsilon^\nu(\vec{0}, \sigma) = u^\mu(\vec{p}, \sigma) \end{cases}$$

Consequently, also at non-zero momentum we can solve for $\varepsilon^\mu(\vec{p}, \sigma) \equiv u^\mu(\vec{p}, \sigma)$ and then take $v^\mu(\vec{p}, \sigma) = (\varepsilon^\mu(\vec{p}, \sigma))^*$.

↪ We thus just need to compute

$$\varepsilon^\mu(\vec{p}, \sigma) = \Lambda_B(\eta, \hat{p})^\mu{}_\nu \varepsilon^\nu(\vec{0}, \sigma)$$

[p. 11****] Example: consider a boost along the $\hat{z}$ direction.

$$\varepsilon^\mu(|\vec{p}|\hat{z}, \sigma) = \begin{pmatrix} E_p/M & 0 & 0 & |\vec{p}|/M \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ |\vec{p}|/M & 0 & 0 & E_p/M \end{pmatrix}^\mu{}_\nu \varepsilon^\nu(\vec{0}, \sigma)$$

■ The polarization vectors with $\sigma = \pm 1$ do not have any component along the time and $\hat{z}$ directions, and they are not affected by the boost.

■ The polarization along $\sigma = 0$ changes.

We thus get

$$\varepsilon^\mu(|\vec{p}|\hat{z}, -1) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -i \\ 0 \end{pmatrix}, \quad \varepsilon^\mu(|\vec{p}|\hat{z}, +1) = -\frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ i \\ 0 \end{pmatrix} \quad \text{“transverse polarizations”}$$

$$\varepsilon^\mu(|\vec{p}|\hat{z}, 0) = \begin{pmatrix} |\vec{p}|/M \\ 0 \\ 0 \\ E_p/M \end{pmatrix} \quad \text{longitudinal polarization.}$$

Added explanation (not in the original notes)

The longitudinal polarization is where the trouble hides. Transverse polarizations are boost-inert (they lie in the plane orthogonal to the boost axis); the longitudinal one is dragged along and *grows without bound*, $\varepsilon^\mu \sim |\vec{p}|/M$ as $|\vec{p}| \rightarrow \infty$. Two consequences worth keeping in mind: (i) transversality is manifestly preserved, $p \cdot \varepsilon(|\vec{p}|\hat{z}, 0) = E_p |\vec{p}|/M - |\vec{p}| E_p/M = 0$ ✓, as it must be since both p and ε were boosted together; (ii) the $1/M$ growth is singular as $M \rightarrow 0$ — the massless limit of a massive vector is not smooth in the longitudinal channel, which is the technical face of the fact that a photon has only two polarizations. This growth is also the origin of the “bad” high-energy behaviour of massive vector-boson scattering in the Standard Model, cured

by the Higgs mechanism — the equivalence theorem states that at high energy a longitudinally polarized W or Z behaves like the corresponding scalar. The ultra-relativistic form derived on [p. 14****] below, $\varepsilon^\mu(\vec{p}, 0) \approx p^\mu/M$, is the precise statement of that growth.

7 The helicity basis [p. 11****–13****]

All our discussion so far was based on the spin-polarization basis. It is of course possible [p. 12****] to formulate the problem in the helicity basis:

$$\Phi^A(x) = \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\kappa u^A(m, \vec{p}, \lambda) e^{-ip \cdot x} a(m, \vec{p}, \lambda) + \eta v^A(m, \vec{p}, \lambda) e^{ip \cdot x} a^\dagger(m, \vec{p}, \lambda) \right]$$

the coefficient functions are now written as functions of the helicity;

annihilation operators now create and destroy particles with definite helicity.

How do we compute $u^A(m, \vec{p}, \lambda)$? Let us remind that the physical states with fixed helicity are defined as follows:

$$\begin{cases} p^\mu = H(\hat{p}, |\vec{p}|/M)^\mu{}_\nu p_{\text{ref}}^\nu & \text{in momentum space} \\ |M, \vec{p}, s, \lambda\rangle = U_H(\hat{p}, |\vec{p}|/M) |M, \vec{0}, s, \sigma\rangle & \text{for physical states} \end{cases}$$

▷ same transformation as H , but “seen” from the representation acting on states.

Not surprisingly, the following transformations hold:

$$\begin{aligned} u^A(m, \vec{p}, \lambda) &= D(H(\hat{p}, |\vec{p}|/M))^A{}_B u^B(m, \vec{0}, \sigma) \\ v^A(m, \vec{p}, \lambda) &= D(H(\hat{p}, |\vec{p}|/M))^A{}_B v^B(m, \vec{0}, \sigma) \end{aligned}$$

▷ same transformation, but seen from the representation acting on the fields.

[p. 13****] Let us compute in our case:

$$\begin{aligned} \varepsilon^\mu(\vec{p}, \lambda) &= H(\hat{p}, |\vec{p}|/M)^\mu{}_\nu \varepsilon^\nu(\vec{0}, \sigma) = \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\rho \underbrace{\Lambda_B(\hat{z}, |\vec{p}|/M)^\rho{}_\nu \varepsilon^\nu(\vec{0}, \sigma)}_{\text{computed in the previous section}} \\ &= \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\nu \varepsilon^\nu(|\vec{p}|\hat{z}, \sigma) \end{aligned}$$

$$\varepsilon^\mu(\vec{p}, \lambda) = \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\nu \varepsilon^\nu(|\vec{p}|\hat{z}, \sigma)$$

▷ here $\lambda = 0, \pm 1$ on the left corresponds to $\sigma = 0, \pm 1$ on the right.

Let us apply to the case $\lambda = 0$:

$$\varepsilon^\mu(\vec{p}, \lambda = 0) = \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\nu \varepsilon^\nu(|\vec{p}|\hat{z}, \sigma = 0) = \Lambda_R(\hat{z} \rightarrow \hat{p})^\mu{}_\nu \begin{pmatrix} |\vec{p}|/M \\ (E_p/M) \hat{z} \end{pmatrix} = \begin{pmatrix} |\vec{p}|/M \\ (E_p/M) \hat{p} \end{pmatrix}.$$

We find

$$\varepsilon^\mu(\vec{p}, \lambda = 0) = \begin{pmatrix} |\vec{p}|/M \\ \frac{E_p}{M} \hat{p} \end{pmatrix} \quad \begin{array}{l} \text{longitudinal polarization vector} \\ \text{in the helicity basis for generic} \\ \hat{p} \end{array}$$

Consider now the ultra-relativistic limit $|\vec{p}| \gg M$. [p. 14****] In this limit $E_p \approx |\vec{p}|$ and the four-momentum p^μ reads

$$p^\mu = \begin{pmatrix} E_p \\ |\vec{p}| \hat{p} \end{pmatrix} \approx |\vec{p}| \begin{pmatrix} 1 \\ \hat{p} \end{pmatrix}.$$

The longitudinal polarization becomes

$$\varepsilon^\mu(\vec{p}, \lambda = 0) = \begin{pmatrix} |\vec{p}|/M \\ (E_p/M) \hat{p} \end{pmatrix} \xrightarrow{\text{UR}} \frac{|\vec{p}|}{M} \begin{pmatrix} 1 \\ \hat{p} \end{pmatrix} = \frac{1}{M} p^\mu$$

$$\varepsilon^\mu(\vec{p}, \lambda = 0) \stackrel{\text{UR}}{\approx} \frac{p^\mu}{M} \quad \begin{array}{l} \text{longitudinal polarization vector in} \\ \text{the ultrarelativistic limit} \end{array}$$

Added explanation (not in the original notes)

Why the rotation $\Lambda_R(\hat{z} \rightarrow \hat{p})$ does the whole job. The standard transformation $H = \Lambda_R(\hat{z} \rightarrow \hat{p})\Lambda_B(\eta, \hat{z})$ of the helicity basis (Lecture 4) differs from the direct boost only by the final rotation, and that rotation acts on the polarization vectors exactly as it acts on any three-vector: it carries $\hat{z}$ into $\hat{p}$ and rotates the transverse pair along with it. That is why, in the helicity basis, λ literally is the spin projection *along the direction of motion* — the transverse polarizations stay orthogonal to $\hat{p}$ and the longitudinal one stays parallel to it, for every $\vec{p}$. Note the internal consistency: at $|\vec{p}| \gg M$, transversality $p \cdot \varepsilon = 0$ is still exact, yet $\varepsilon^\mu \rightarrow p^\mu/M$ — no contradiction, because $p \cdot p = M^2$ is negligible *relative* to the $|\vec{p}|^2/M^2$ scale of the vector itself: the longitudinal polarization becomes proportional to p^μ only up to corrections of relative order $M^2/|\vec{p}|^2$, which is exactly what carries the transversality.

8 Sum over physical polarizations [p. 14****–16****]

Consider the field in the helicity basis:

$$V^\mu(x) = \sum_\lambda \int \frac{d^3\vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left[\kappa \varepsilon^\mu(\vec{p}, \lambda) e^{-ip \cdot x} a(\vec{p}, \lambda) + \eta \varepsilon^\mu(\vec{p}, \lambda)^* e^{ip \cdot x} a^\dagger(\vec{p}, \lambda) \right]$$

We compute the quantity $\sum_{\lambda=\pm 1, 0} \varepsilon^\mu(\vec{p}, \lambda) \varepsilon^\nu(\vec{p}, \lambda)^*$. We use the following strategy.

i) We make an ansatz by writing

$$\sum_\lambda \varepsilon^\mu(\vec{p}, \lambda) \varepsilon^\nu(\vec{p}, \lambda)^* = A g^{\mu\nu} + B p^\mu p^\nu$$

▷ it must be a tensor under Lorentz: the only possibilities are, [p. 15****] using $g^{\mu\nu}$ and $p^\mu p^\nu$.

ii) We use the transversality condition $p_\mu \varepsilon^\mu(\vec{p}, \lambda) = 0$:

$$p_\mu (A g^{\mu\nu} + B p^\mu p^\nu) = A p^\nu + B p^2 p^\nu = (A + B p^2) p^\nu \stackrel{!}{=} 0$$

$$\rightsquigarrow B = -\frac{A}{p^2} = -\frac{A}{M^2}$$

(since we consider physical polarizations, we have $p^2 = M^2$). We arrive at

$$\boxed{\sum_{\lambda} \varepsilon^{\mu}(\vec{p}, \lambda) \varepsilon^{\nu}(\vec{p}, \lambda)^* = A \left(g^{\mu\nu} - \frac{p^{\mu} p^{\nu}}{M^2} \right)}$$

iii) We compute “ A ” by matching the two sides for a specific value of $p \rightsquigarrow$ we consider the rest frame, in which $p^{\mu} = (M, \vec{0})$. We take $\mu = \nu = 1$:

$$\sum_{\lambda} \varepsilon^1(\vec{0}, \lambda) \varepsilon^1(\vec{0}, \lambda)^* = A \left(g^{11} - \frac{p^1 p^1}{M^2} \right) = A(-1 - 0) = -A.$$

From the explicit expressions at zero momentum:

$$\underbrace{\left(-\frac{1}{\sqrt{2}} \right) \left(-\frac{1}{\sqrt{2}} \right)}_{\text{from } \lambda=1} + \underbrace{\left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right)}_{\text{from } \lambda=-1} = 1.$$

[p. 16****] We thus find $A = -1$ and

$$\sum_{\lambda=0, \pm 1} \varepsilon^{\mu}(\vec{p}, \lambda) \varepsilon^{\nu}(\vec{p}, \lambda)^* = -g^{\mu\nu} + \frac{p^{\mu} p^{\nu}}{M^2} \quad \begin{array}{l} \text{polarization sum, massive} \\ \text{spin-1 particle} \end{array}$$

Added explanation (not in the original notes)

Reading the polarization sum. The tensor $-g^{\mu\nu} + p^{\mu} p^{\nu}/M^2$ is (minus) the projector onto the three-dimensional subspace orthogonal to p^{μ} : one checks $p_{\mu}(-g^{\mu\nu} + p^{\mu} p^{\nu}/M^2) = -p^{\nu} + p^{\nu} = 0$ ✓ and that its trace is $-4 + p^2/M^2 = -3$ — exactly the number of physical polarizations ✓. The $\lambda = 0$ contribution is what supplies the $p^{\mu} p^{\nu}/M^2$ piece, so this term again carries the $1/M^2$ that blows up in the massless limit; in a massive vector propagator, $\frac{-i(g^{\mu\nu} - p^{\mu} p^{\nu}/M^2)}{p^2 - M^2 + i\epsilon}$, it is the same term that makes the naive power counting of the theory worse than for a photon (whose propagator has no such piece thanks to gauge invariance). Two remarks on the derivation: (i) the ansatz is justified because the left-hand side is a Lorentz tensor built only out of p^{μ} and the invariant tensors — $\epsilon^{\mu\nu\rho\sigma}$ cannot appear since the left side is symmetric under $\mu \leftrightarrow \nu$ combined with conjugation; (ii) the matching in step iii) may be done in the rest frame with a single component precisely because the tensorial form has already been fixed — one number suffices. Note also that the Condon–Shortley sign drops out, as promised, since it appears squared.

9 Microcausality and the connection between spin and statistics

[p. 16****–18****]

We impose the condition

$$[V^{\mu}(x), V^{\nu}(y)]_{\pm} = 0, \quad (x - y)^2 < 0$$

(▷ leave open the possibility that the fields either commute or anti-commute).

$$[V^{\mu}(x), V^{\nu}(y)]_{\pm} = \sum_{\lambda} \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \kappa_{\eta} \varepsilon^{\mu}(\vec{p}, \lambda) \varepsilon^{\nu}(\vec{p}, \lambda)^* \left[e^{-ip \cdot (x-y)} \pm e^{ip \cdot (x-y)} \right]$$

$$\stackrel{\text{pol. sum}}{=} \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \kappa \eta \left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{M^2} \right) \left[e^{-ip \cdot (x-y)} \pm e^{ip \cdot (x-y)} \right] \stackrel{(*)}{=}$$

We now trade powers of p with derivatives acting on $e^{\mp ip \cdot (x-y)}$. In particular, we note that [p. 17****]

$$\frac{\partial^2}{\partial x^\mu \partial x^\nu} e^{\mp ip \cdot (x-y)} = (\mp i p_\mu)(\mp i p_\nu) e^{\mp ip \cdot (x-y)} = -p_\mu p_\nu e^{\mp ip \cdot (x-y)}$$

$$\rightsquigarrow \boxed{p^\mu p^\nu e^{\mp ip \cdot (x-y)} = -\frac{\partial^2}{\partial x_\mu \partial x_\nu} e^{\mp ip \cdot (x-y)}}$$

and

$$\stackrel{(*)}{=} \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \kappa \eta \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \left[e^{-ip \cdot (x-y)} \pm e^{ip \cdot (x-y)} \right]$$

$$= \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \kappa \eta \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} \left[e^{-ip \cdot (x-y)} \pm e^{ip \cdot (x-y)} \right]$$

Using

$$\boxed{\Delta_+(x-y) \equiv \int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} e^{-ip \cdot (x-y)}}$$

we arrive at:

$$[V^\mu(x), V^\nu(y)]_\pm = \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \kappa \eta \left[\Delta_+(x-y) \pm \Delta_+(y-x) \right]$$

► This result is valid for any x and y . We now restrict to $(x-y)^2 < 0$.

Added explanation (not in the original notes)

(Transcription note.) On [p. 17****] the factor $1/M^2$ accompanying the double derivative is dropped in the handwritten manipulations, although it is present in the polarization sum being substituted; it has been restored above, since $p^\mu p^\nu / M^2 \rightarrow -(1/M^2) \partial^2 / \partial x_\mu \partial x_\nu$. It plays no role in the argument that follows — the differential operator multiplies the whole bracket and cannot change whether it vanishes.

What Δ_+ is. It is the same positive-frequency Wightman function met at the end of Lecture 7: the vacuum two-point function of a scalar field, $\Delta_+(x-y) = \langle 0 | \phi(x) \phi(y) | 0 \rangle$, i.e. the mass-shell integral of $e^{-ip \cdot (x-y)}$ with $p^0 = E_p > 0$. The second term is its $x \leftrightarrow y$ mirror, which is $\Delta_+(y-x) = \Delta_+(x-y)^*$. Both are nonzero outside the light cone (they decay like $e^{-M\sqrt{-(x-y)^2}}$), which is why neither term alone can satisfy causality — exactly the impasse flagged in Lecture 7 ([p. 25****]). The whole burden of the argument therefore falls on the *cancellation* between them, which is the content of the next page.

[p. 18****] The function $\Delta_+(x-y)$ enjoys the property that

$$\Delta_+(x-y) = \Delta_+(y-x) \quad \text{if } (x-y)^2 < 0.$$

Consequently, if $(x - y)^2 < 0$, we write

$$[V^\mu(x), V^\nu(y)]_\pm = \left(-g^{\mu\nu} - \frac{1}{M^2} \frac{\partial^2}{\partial x_\mu \partial x_\nu} \right) \kappa \eta \underbrace{\left[\Delta_+(x - y) \pm \Delta_+(x - y) \right]}_{\substack{\text{the only possibility to get zero} \\ \text{is to pick the minus sign}}} \stackrel{!}{=} 0$$

► Annihilation and creation operators must obey commutation relations in order to have a field theory description of massive spin-1 particles compatible with relativistic causality.

► At this stage, κ and η are still arbitrary complex numbers. However, we still have to impose the microcausality condition

$$[V^\mu(x), V^\nu(y)^\dagger] = 0 \quad \text{if } (x - y)^2 < 0.$$

Added explanation (not in the original notes)

This is the spin–statistics theorem, in the spin-1 case. Recall the logic skeleton: the “ $\pm$ ” in $[\cdot, \cdot]_\pm$ is inherited from the (anti)commutation relations postulated for a and $a^\dagger$ (Lecture 6), which were *assumed* there to correlate with the spin. Here nothing was assumed — and the requirement of causality has just selected the commutator. Since the polarization sum of a spin-1 particle is symmetric under $\mu \leftrightarrow \nu$ together with $p \rightarrow p$, the two terms $\Delta_+(x - y)$ and $\Delta_+(y - x)$ enter with the *same* sign, and outside the light cone they are equal; only the minus sign in the bracket can annihilate them. Running the same computation for a spin- $\frac{1}{2}$ field gives the opposite outcome: the analogous “polarization sum” (the Dirac spin sum $\sum_s u \bar{u} = p_\mu \gamma^\mu + m$) is *odd* in p^μ , and the momentum integral produces a relative minus between the two terms, so causality then demands the *anticommutator*. That is the general statement: integer spin $\rightarrow$ commutators (Bose), half-integer spin $\rightarrow$ anticommutators (Fermi).

Why the story is not over. The vanishing of $[V^\mu(x), V^\nu(y)]$ is only half of the microcausality requirement stated back in Lecture 7 ([p. 26***]): the mixed bracket $[V^\mu(x), V^\nu(y)^\dagger]$ must vanish too. That second condition is what will constrain the still-free constants: it involves $|\kappa|^2$ and $|\eta|^2$ (rather than the product $\kappa\eta$), so it will fix $|\kappa| = |\eta|$ — the statement that particles and antiparticles must be created and destroyed with equal weight, i.e. the existence of antiparticles with the same mass and spin. Presumably the next lecture picks up exactly there.

(End of Lecture 9 — the notes continue with the second microcausality condition and the determination of κ and η .)